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SUPERSEDING
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FEDERAL SPECIFICATION

BAG, SOILED CLOTHES (WATER SOLUBLE PLASTIC)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for water soluble soiled clothes bag.

1.2 Classification.

1.2.1 Types and sizes. Water soluble soiled clothes bags covered by this specification shall be of the following types and sizes, as specified (see 6.2).

Type I - 55°F and above water soluble film.

Size 1 - 26 inches wide by 33 inches high.
Size 2 - 28 inches wide by 39 inches high.
Size 3 - 36 inches wide by 39 inches high.

Type II - 140°F and above water soluble film.

Size 1 - 26 inches wide by 33 inches high.
Size 2 - 28 inches wide by 39 inches high.
Size 3 - 36 inches wide by 39 inches high.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Philadelphia, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Houston, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

FSC 8465

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Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standard:

D 1922 - Propagation Tear Resistance of Plastic Film and Thin Sheeting.

(Copies may be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Film. The film shall be water soluble polyvinyl alcohol film (see 3.1.1.1 and 3.1.1.2). At the option of the supplier, the film shall be either glossy finished on one side and matte finished on the other side or be glossy finished on both sides. The film shall conform to the requirements specified hereinafter.

3.1.1.1 Type I bag. The film for type I bag shall begin to disintegrate within 40 seconds in tap water having a temperature of 40° to 50°F when tested as specified in 4.4.2.2. Also, there shall be no more than 1 percent residue remaining after agitating 100 seconds in distilled water having a temperature of 55° ± 5°F when tested as specified in 4.4.1.3. In addition, the film shall have the minimum physical properties specified in 4.4.1.2 and shall melt but not burst into flame when tested as specified in 4.4.2.3 nor shall the bag rupture when tested as specified in paragraph 4.4.2.1.

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TABLE I. Physical properties - type I film

Property	Requirement 50 percent R. H.*/	Requirement 95 percent R. H.*/	Test Method
Internal Tear Resistance, gm/ml (minimum)	400	250	ASTM D 1922
Thickness, inches	0.0013 + 0.0004	----	----------
Solubility (dissolves)	55° + 5°F	----	Para. 4.4.1.3

*/ Relative humidity + 2 percent.

3.1.1.2 Type II bag. The film for type II bag shall begin to disintegrate within 50 seconds in tap water having a temperature of 110° to 120°F when tested as specified in 4.4.2.2. Also, there shall not be more than 2 percent residue remaining after agitating for 300 seconds in distilled water having a temperature of 140° plus 5°F when tested as specified in 4.4.1.3. In addition, the film shall have the minimum physical properties specified in table II when tested as specified in 4.4.1.2 and shall melt but not burst into flame when tested as specified in 4.4.2.3, nor shall the bag rupture when tested as specified in paragraph 4.4.2.1.

TABLE II. Physical properties - type II.

Property	Requirement	Test Method
Tear Resistance, (minimum)	500 Type II (50 + 2% R.H.)	ASTM D 1922
Thickness, inches	0.0013 + 0.0004	----------
Solubility (dissolves)	140° + 5°F - Type II	Para. 4.4.1.3

3.1.2 Closure tie film. Type I film (see 3.1.1.1) shall be used for the closure tie for the types I and II.

3.2 Color. The color of the film used for the bags and tie closures shall be the manufacturer's standard commercial color.

3.3 Design. The bag shall be a standard commercial flat, rectangular shape design, open across one end of its width. Bags shall be designed to open without difficulty. The closure tie shall be attached to the bag by the method normally used by the supplier.

3.4 Type I bag. The type I bag shall be made from the film specified in 3.1.1.1 and shall meet the dimensional requirements in table III as specified (see 6.2).

3.4.1 Type II bag. The type II bag shall be made from the film specified in 3.1.1.2 and shall meet the dimensional requirements in table III as specified (see 6.2).

3.4.2 Closure tie. The closure tie shall be made from a single thickness of type I film. It shall meet the dimensional requirements specified in table III.

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TABLE III. Dimensional requirements (inches)

Requirements	Bag sizes (inches) 1	2	3	Tolerance
Bag:				
Width (seam to seam)	26	28	36	± 1/4
Length (top edge to bottom seam)	33	39	39	± 1/4
Closure tie:				
Width	3/4	3/4	3/4	± 1/2
Length	38	39	39	± 1 1/
Location (edge bag opening to closure tie)	5	5	5	± 2

1/ Tie closures caught in side seams shall be of sufficient length to allow the bag to lay flat.

3.4.3 Seams. The seams shall be sealed and trimmed in accordance with standard commercial practice.

3.5 Instructions. A card or sheet with instructions describing the proper use of the bag shall be included with each intermediate package of bags. The card or sheet shall be marked with the applicable type and size, and month and year packed (see 6.3). The card or sheet shall be packed in such a manner that the information is visible through the intermediate pack outer wrap.

3.6 Workmanship. The finished water soluble bags shall be free from any weaking defects such as tears or pinholes. The bags shall be free from foreign matter and of irregular creases. The bags shall be fabricated with the matte side out and with closure ties securely attached. The bags shall be properly trimmed and neatly folded with the full width of one end open.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling for inspection and acceptance. Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Film physical properties. The film shall be tested for the minimum physical properties specified in either table I or II, whichever is applicable, in accordance with 4.4.1. Test shall be made on the first lot offered for inspection, and on every 5,000 pounds film manufactured thereafter, or once every three months whichever is more frequent.

4.2.2 End item inspection. The end item shall be inspected as specified in 4.2.2.1 through 4.2.2.3. The sample unit shall be one completely fabricated bag.

4.2.2.1 Visual examination. Examination shall be made of the end item for defects in table IV. The inspection level shall be level II with an AQL of 4.0 expressed in terms of defects per hundred units.

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TABLE IV. Visual defects

Examine	Defects
Finish	Film not free of pinholes or particles of foreign matter. Areas with irregular creases.
Design, Construction, and Workmanship	Bags difficult to open. Seams not properly sealed. Closure tie not securely attached to bag. Bag not neatly and properly trimmed. Not open across the full width of one end.

4.2.2.2 Dimensional examination. Inspection shall be made of the end item for compliance with table III. Any deviation from specified requirements shall constitute a defect. The inspection level shall be S-2 with an AQL of 4.0 expressed in terms of defects per hundred units.

4.2.2.3 Testing of the end item. The end item shall be tested as specified in 4.4.2. The inspection level shall be S-2 with an AQL of 4.0 expressed in terms of defects per hundred units.

4.3.3 Inspection of preparation for delivery. An inspection shall be made to determine whether the packaging, packing, and marking comply with the requirements in section 5 of this specification. Defects shall be scored in accordance with table V. For examination of interior packaging, the sample unit shall be one shipping container fully prepared for delivery, selected at random just prior to the closing operations. Sampling shall be in accordance with MIL-STD-105. The lot size shall be the number of shipping containers in the end item inspection lot. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The inspection level shall be S-2 with an AQL of 4.0 defects per hundred units.

TABLE V. Classification of preparation for delivery defects

Examine	Defects
Markings (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping or distortion of container.
Contents (exterior and interior container)	Number per container is more or less than required. Use instructions missing or not visible; not marked with applicable type and size; or month and year packed.

4.4 Test.

4.4.1 Film. The film used to fabricate the bags (see 3.1.1.1 and 3.1.1.2) shall be tested in accordance with either the specified ASTM Test Methods, or test specified herein as applicable. Samples of the film shall be selected and cut into the required test specimens as specified in the applicable test or test method. The results obtained from each test shall be averaged. The failure of any test requirement shall be cause for rejection of the lot of film.

4.4.1.1 Conditioning. The test specimens shall be conditioned in an atmospheric condition of 23° ± 1°C (73.4 ± 1.8°F) and 50 ± 5 percent relative humidity for 24 hours, with testing proceeding immediately afterwards and in the same atmospheric conditions. The average values shall be reported.

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4.4.1.2 Tear resistance. Test specimens shall be cut to constant-ra and tested in accordance with ASTM Method D 1922.

4.4.1.3 Solubility. The film shall be tested as specified in 4.4.1. 4.4.1.3.2 to determine that film dissolves at the temperature and within specified in 3.1.1.1 and 3.1.1.2, whichever is applicable.

4.4.1.3.1 Equipment and material.

(a) Wrist-action shaker. Burrell Cat. No. 75-775 BB or equal.
(b) Oven, air circulating.
(c) Balance, analytical.
(d) Timer.
(e) Thermometer, plus 20° to 220°F in 2° divisions maximum.
(f) 100 Mesh Wire Screens.
(g) Liter Erlenmeyer Flask w/rubber stop.
(h) Petri dish.
(i) Desiccator.
(j) Template 2-1/2 by 5 + 1/16 inches.
(k) Actone (commercial grade).
(l) Distilled water.

4.4.1.3.2 Procedure. Three samples of the applicable film shall be and cut using the template as a guide and conditioned as specified in 4. for each test. After conditioning the samples shall be individually wei the nearest 0.1 milligram and their individual weight recorded.

The filter screens shall be pre-dried in an air circulating oven at 250° for a period of not less than 30 minutes. At the end of the drying peri shall be placed in a desiccator to cool. After cooling the screens shal weighed individually to the nearest 0.1 milligrams and their individual recorded.

Fill liter flask with 500 plus 0 minus 20 milliliters of distilled water of the specified temperature. Place flask on shaker and set degree of a at two. Cut a weighed film sample into squares of 1/2 inch or less in s drop into the flask. Place the stopper securely into the flask and agit the time specified in 3.1.1.1 or 3.1.1.2, whichever is applicable.

NOTE: Be sure cut film pieces do not cling to sides of flask.

Immediately after agitation, filter the contents of the flask through a screen. Rinse the flask thoroughly with cold water and filter through t screen to insure complete transfer of any residue remaining in the flask screen.

After all water has drained through the filter, immerse it in a petri di filled with acetone for a period of not less than 2 minutes. Then remov screen and dry in the oven for a period of not less than 30 minutes at 2 10°F. After drying, cool in a desiccator and when it has cooled, weigh nearest 0.1 milligram and record.

The percent of residue remaining shall be calculated by the formula perc residue =

$$\frac{A-B \times 100}{C}$$

Where: A = Weight of screen plus residue.
B = Weight of screen.
C = Weight of film sample.

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The test shall be repeated using the other two weighed samples. The results of each test shall comply with 3.1.1.1 or 3.1.1.2, whichever is applicable. Failure of any of the test samples shall be cause for rejection.

NOTE: Use same arm of the shaker for test reproducibility.

4.4.2 End item test. End item test shall be conducted in accordance with 4.4.2.1 through 4.4.2.4 using completed bags that meet the dimensional requirements specified in table III.

4.4.2.1 Drop test conditioning. The bags shall be preconditioned and tested in a controlled atmosphere of 73.4° ± 1.8°F and 50 ± 5 percent R.H. The bags shall be pre-conditioned 4 hours ± 15 minutes prior to filling and testing.

4.4.2.1.1 Drop test. The type 1 or 2, size 1, 2, or 3 bags shall be loaded with the following amounts of dry linen prior to testing:

Size 1 - 10 to 15 pounds.
Size 2 - 20 to 25 pounds.
Size 3 - 30 to 35 pounds.

The term dry linen is defined as linen which has been dried at ambient room temperature and humidity.

The testing sequence of the faces, edges and butts (top and bottom) shall be identified as follows: Facing the front of the bag (wide side where tie closure is knotted) with top up this side shall be designated as the number 1 test position, the right side edge as 2, the rear side (side opposite tie closure knot) as 3, the left side as 4, the bottom as 5, and the top as 6. Before each drop the contents shall be uniformly distributed throughout the bag.

The bag to be tested shall be loaded with the specified amount of linen and prior to closing with its own tie closure all air shall be expelled from inside the bag. The bag shall again be checked after closing to assure no air is trapped inside prior to testing.

The bag shall be positioned in a suitable quick-release mechanism over a solid, smooth surface of sufficient mass to absorb all shock without deflection. The drops shall be made from a height of 8 feet plus 6 inches, as measured from the underside of the bag, in numerical test sequence cited, with the face on which the bag is to fall parallel to the floor. The drops shall be vertical, free, and unobstructed. After each drop the bag shall be examined for compliance with either 3.1.1.1 or 3.1.1.2, whichever is applicable. Failure to comply shall be cause for rejection.

4.4.2.2 Film disintegration. A specimen of film, using the template, shall be cut from the same bag the seam test specimens were obtained. In addition, a specimen the full width of the tie closure and 1-1/4 to 1-1/2 inches long shall be cut from the bag's tie closure.

A petri dish shall be filled approximately 3/4 full of water of the specified temperature. A single specimen shall be carefully placed upon the surface of the water in such a manner to prevent the film folding back upon itself. The specimen shall be visually observed for signs of disintegration in compliance with 3.1.1.1 or 3.1.1.2, whichever is applicable. Failure to show signs of disintegration within the time and at the water temperature specified shall be cause for rejection.

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4.4.2.3 Film flammability. A specimen of film, using the template, shall be cut from the same bag the seam test specimen was obtained. The specimen shall be clamped smooth and taut in a suitable fixture. A soldering iron (chisel tip. w/heat range 700° - 850°F) heated to operating temperature shall be brought in contact with film and moved along its face to determine compliance with 3.1.1.1 or 3.1.1.2, whichever is applicable.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or commercial, as specified (see 6.2).

5.1.1 Level A. Twenty-five bags shall be enclosed in a polyethylene bag or wrap, a minimum of 1.5 mils thick and waterproof sealed.

5.1.2 Commercial. The bags shall be packaged in accordance with normal commercial practice. The complete package shall be designed to protect the item against damage during shipment, handling and storage.

5.2 Packing. Packing shall be level A or commercial, as specified (see 6.2).

5.2.1 Level A. Four unit packages (100 bags), packaged as specified in 5.1.1, shall be packed in a box made from weather-resistant fiberboard having a bursting strength of not less than 275 pounds per square inch. The box flaps shall be secured with water-resistant adhesive applied to the entire area of contact between the flap or with 3 inch wide water-resistant tape applied to the full length of the outer seams and extending over the ends not less than 3 inches. Alternatively, cleated plywood, wirebound, or nailed wood boxes shall be acceptable shipping containers when lined with a waterproof barrier material. The edges of the barrier material shall be sealed with waterproof tape or adhesive.

5.2.2 Commercial. The bags, packaged as specified in 5.1.1, shall be packed in fiberboard boxes to insure safe delivery at destination, to provide for safe redistribution by the initial receiving activity, and shall be acceptable by common carrier under National Motor Freight Classification or Uniform Freight Classification.

5.3 Unitization. When shipments to government depots are full car or truckload the shipping containers shall be unitized for shipment and handling in accordance with normal commercial practice. The unitized load shall not exceed 2,500 pounds in weight, 63 inches in height, 56 inches in length, and 45 inches in width.

5.4 Marking. Marking shall be in accordance with Fed. Std. No. 123.

6. NOTES

6.1 Intended use. The water soluble bags covered by this specification are intended to help avoid the spread of contamination by suppressing airborne microorganisms and by reducing human contact with soiled and contaminated hospital, quarantine station, etc. linen.

6.1.1 Type I. Used for washing woolen blankets, colored fabrics, etc. requiring warm water wash cycle.

6.1.2 Types II and III. Used for washing linen, clothing, etc. that require a hot water wash cycle of 140°F and 160°F respectively and above. The bag burst's during the initial cold flush cycle releasing the contents.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type and size (see 1.2).
(c) Applicable levels of packaging and packing (see 5.1 and 5.2).

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6.3 Based on available technical information, it is suggested that users observe the following:

(a) Over-wrap all damp and wet items with dry items before placing in bag. This assures the bag will remain intact while in transit to the laundry room. CAUTION: Do not place overly wet items in bag.
(b) Expel all air from bag before tieing shut. CAUTION: Avoid expelling air in face.
(c) Do not knot top of bag, tie top securely with tie closure only.
(d) Do not handle bag with wet hands, or allow bag to come in contact with water or other damp or wet materials.
(e) Suggest bags known or suspected of containing contaminated linen be placed in cloth, polyethylene, or other suitable type bag to prevent tearing, puncturing, or spacing the soluble bag and truck to laundry room. Remove from outer bag and place soluble bag with soiled items in wash wheel. Wash or dispose of protective bag as directed.
(f) Perform both cold and hot water flush cycles before adding washing chemicals.
(g) Use type II bag for laundry that require hot wash cycle of 140°F or more. Use type III bag for laundry that require hot wash cycle of 160°F or greater. Type I bag may be used for laundry requiring either warm or hot wash cycle.
(h) Do not exceed rated machine capacity and maintain proper water levels.
(i) Follow instructions furnished with each intermediate package of bags.
(j) Bags should be stored at 45° minus 85°F and at relative humidities between 20 and 70 percent and should be used within 12 months. Keep bags in protective over wrap until used. CAUTION: The bag film has the ability to acquire or lose moisture. A too great a loss or gain can affect the bag strength.

MILITARY INTEREST:

None

Preparing activity:

GSA - FSS

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